Detection-Probability-Driven Resource Allocation for ISAC with Joint Power and Waveform Co-Design

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Abstract. This paper replaces the usual SNR-based sensing metric in an integrated sensing and communication (ISAC) power-allocation problem with detection probability, so that waveform shaping becomes a direct design variable rather than a side detail. We study a weighted rate–detection formulation over a time-varying OFDM channel and optimize both power and a normalized waveform profile. The experiments use a `UMi_NLOS` scenario with false-alarm probability $P_{\mathrm{fa}} = 10^{-3}$, which defines the detection regime for the reported P_d values. In the Pareto study, Grey Wolf Optimizer (GWO) spans the tradeoff from a sensing-focused point with $P_d = 0.892$ at 10.70 bps/Hz to a communication-focused point with 22.76 bps/Hz and $P_d = 0.212$. Compared with water-filling, which reaches 22.50 bps/Hz but only $P_d = 0.114$, the proposed formulation keeps nearly the same rate while almost doubling detection probability at the rate-focused end, and it reaches a much stronger sensing point at the opposite end. This suggests that detection-aware waveform co-optimization gives a more useful ISAC tradeoff than power-only allocation.

Keywords: ISAC · detection probability · waveform co-optimization · UMi_NLOS · Pareto analysis

1 Introduction

Integrated sensing and communication (ISAC) tries to use the same transmission resources for both data transfer and sensing. For a detection-oriented sensing task, probability of detection is usually a better measure than sensing SNR alone. This paper therefore studies a detection-probability-centered ISAC problem instead of an SNR-centered one.

Our work aims to answer the following question: in a manageable and realistic `UMi_NLOS` environment, which optimization method attains the best sensing-communication tradeoff with detection probability and waveform shaping kept in mind ?

The main contributions of this paper are:

- expansion of the feasible design space beyond power-only allocation, allowing the optimizer to steer energy towards sensing or communication-favourable subcarriers,

– a multi-algorithm benchmark on a realistic channel, providing a fair and reproducible comparison of meta-heuristic search behavior, and,
– a Pareto analysis showing the value of waveform co-optimization,

2 Related Work and Scope

ISAC was firstly studied from different perspectives, including its limits, the waveforms and its system-level tradeoffs [1,2]. A consistent lesson from that literature is that the sensing metric must match the task. If the task is detection, then probability of detection is more meaningful than a proxy such as sensing SNR.

Waveform design is also a common theme in OFDM-based ISAC. Prior work has shown that waveform shaping can change the sensing–communication balance in a real way [3]. This paper follows that idea, but keeps the model simple enough to focus on optimization behavior rather than on a very detailed radar signal chain.

The channel side of ISAC also matters. Standard channel descriptions such as 3GPP TR 38.901 motivate scenario-based studies instead of generic fading only [4]. That is why this paper uses `UMi_NLOS` and a time-varying channel sequence.

The scope is intentionally limited:

– one target return is modeled on the sensing side,
– clutter and multi-target effects are not modeled,
– the static study uses one representative snapshot,
– the paper compares optimization methods, not detector implementations.

3 System Model

We consider an OFDM system with 32 subcarriers and a time-varying channel sequence of 20 CSI snapshots in the `UMi_NLOS` scenario. The channel includes path loss, shadowing, multipath delay spread, and Doppler variation. Communication and sensing use separate effective channel responses.

The representative snapshot in Figure 1 is clearly frequency selective, which is what makes joint power and waveform shaping useful in this setting.

3.1 Waveform coupling

The waveform profile $\mathbf{w}$ is a normalized per-subcarrier shaping profile. It is not an antenna beamformer because the model does not include array geometry. Instead, $\mathbf{w}$ is a simple spectral weighting vector that changes how strongly each subcarrier contributes to the communication and sensing branches.

We apply waveform shaping by raising each per-subcarrier weight to an exponent. Let $w_{\min} > 0$ be a small floor value used for numerical stability. The communication rate is

$$R(\mathbf{p}, \mathbf{w}) = \sum_{k=1}^{K} \log_2 \left(1 + \frac{p_k \, g_{\mathrm{comm},k} \left(\max(w_k, w_{\min})\right)^{\beta_c}}{N_0} \right), \tag{1}$$

and the sensing SNR proxy is

$$\mathrm{SNR}_{\mathrm{sense}}(\mathbf{p},\mathbf{w}) = \frac{\sum_{k=1}^{K} p_k\, g_{\mathrm{sense},k} \left(\max(w_k, w_{\min})\right)^{\beta_s}}{K\, N_0}. \tag{2}$$

Here N_0 is the (per-subcarrier) noise power, and $g_{\mathrm{comm},k}$ and $g_{\mathrm{sense},k}$ are the effective communication and sensing gains at subcarrier k.

The exponents β_c and β_s control how strongly waveform shaping affects communication and sensing. In our default setting, $\beta_c = 0.15$ and $\beta_s = 1.0$; we also use $w_{\min} = 10^{-6}$. When waveform co-optimization is disabled, $\mathbf{w}$ is fixed to the flat profile $w_k = 1$.

The waveform constraints are:

$$\sum_{k=1}^{K} w_k = K, \qquad w_k \geq w_{\min} \ \ \forall k, \tag{3}$$

and power is constrained by a total budget with optional per-subcarrier caps.

3.2 Detection probability mapping

The sensing side uses a simple single-target, round-trip return model. An effective sensing SNR is first computed via (2), then mapped to probability of detection using a Gaussian-tail approximation (so the paper is fully reproducible).

Let $Q(\cdot)$ be the standard Gaussian tail function, i.e., $Q(x) = 1 - \Phi(x)$. For a fixed false-alarm probability P_{fa}, define the threshold

$$\eta = Q^{-1}(P_{\mathrm{fa}}) = \Phi^{-1}(1 - P_{\mathrm{fa}}), \tag{4}$$

where $\Phi(\cdot)$ is the standard Gaussian CDF. We then apply an integration gain G_{int} (a tunable modeling constant) and compute

$$P_d = Q\Big(\eta - \sqrt{2\, G_{\mathrm{int}}\, \mathrm{SNR}_{\mathrm{sense}}(\mathbf{p},\mathbf{w})}\Big). \tag{5}$$

This form follows standard Gaussian detection models used to relate an effective SNR to a tail probability [9,10]. In the reported experiments we use $P_{\mathrm{fa}} = 10^{-3}$ and $G_{\mathrm{int}} = 0.02$. We treat G_{int} as a modeling knob that controls how quickly P_d rises with sensing SNR; it is not presented as a detector-derived constant.

The objective is

$$J(\mathbf{p},\mathbf{w}) = \alpha R(\mathbf{p},\mathbf{w}) + (1-\alpha)\gamma U_{\mathrm{sense}}, \tag{6}$$

where $U_{\mathrm{sense}} = P_d$ in the default mode and γ is a scaling factor used to keep the sensing term on a comparable numerical scale to the rate term. In all reported results, $\gamma = 10$.

3.3 Physical Interpretation of the Waveform Profile

The waveform profile $\mathbf{w}$ can be read as a "spectrum emphasis" vector. If $\mathbf{w}$ is flat, all subcarriers contribute equally. If $\mathbf{w}$ is non-flat, some subcarriers are emphasized more than others, which lets the optimizer steer the design toward sensing-favorable or communication-favorable frequency regions.

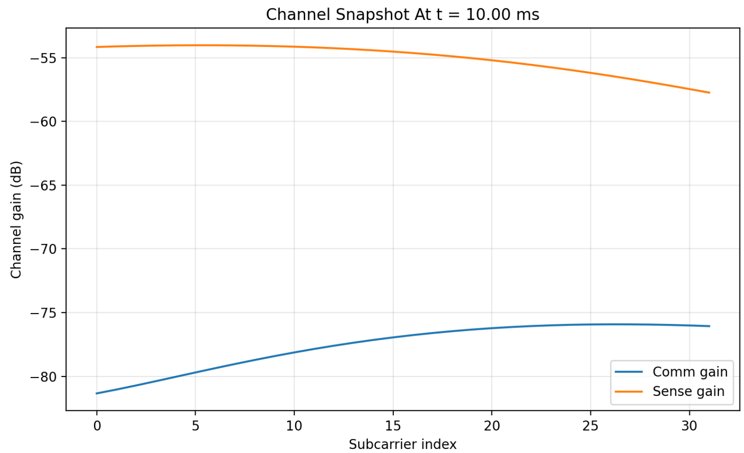

Fig. 1. Representative `UMi_NLOS` channel snapshot used in the static study.

3.4 Scope and Modeling Assumptions

The sensing model is intentionally made to be as simple as possible. It uses a point target only, without any clutter, extended target effects, and also without noise-clutter covariance consideration. This kind of modeling allows for isolation of the optimization characteristics, i.e., how the detection performance relates to waveform design under actual channel effects. Such an approach will distinguish our results from radar receiver engineering considerations.

4 Method and Evaluation Protocol

The same problem instance is given to every algorithm. This keeps the comparison fair and makes the results easier to read. The methods are compared in three ways:

- a single representative snapshot,
- a dynamic sequence of 20 channel updates,
- a Pareto-style sweep over the weighted objective.

These settings are fixed in advance and are not tuned per solver.

4.1 Reproducibility and Statistical Scope

All solvers are run with fixed random seeds, as listed in Table 2. This makes the runs exactly reproducible, but it also limits statistical strength. Since the gaps between the top two solvers (GWO and PSO) are small, we describe GWO as having a slight edge under the fixed-seed setup rather than claiming definitive superiority. A multi-seed study with confidence intervals is a natural next step, but it is outside the scope of this case study.

Table 1. Key system and objective parameters used throughout the case study.

Parameter	Value
Scenario	`UMi_NLOS`
Subcarriers K	32
Total power P_{tot}	10 W
Per-subcarrier cap $p_k \le P_{\max}$	1.25 W
Noise power N_0	10^{-8} W
False-alarm probability P_{fa}	10^{-3}
Integration gain G_{int}	0.02
Waveform floor $w_{\min}$	10^{-6}
Waveform exponents (β_c, β_s)	(0.15, 1.0)
Objective scaling γ	10
Default weight α (static/dynamic)	0.5

Table 2. Solver settings used in the manuscript.

Solver	Main setting	Secondary setting	Seed	Notes
GWO	population=30, iterations=50	mutation $\sigma = 0.02$	11	main weighted solver
PSO	population=30, iterations=50	inertia 0.7, cognitive/social 1.5	17	close competitor
DE	population=30, iterations=50	difference weight 0.8, crossover 0.9	19	evolutionary baseline
SA	iterations=1000	initial $T = 100$, final $T = 0.1$	23	fastest solver
SFO	population=30, iterations=50	regeneration 0.1	31	baseline
POA	population=30, iterations=50	attack probability 0.2	37	baseline
NSGA-II	population=48, generations=60	crossover 0.9, mutation 0.12	29	reference front

4.2 Computational Complexity

The running time of all methods depends mainly on the number of evaluations of the objective functions. Any increase in the size of the population or in the number of iterations results in an increase in the number of evaluations, therefore increasing the running time. The NSGA-II method incurs additional costs by virtue of its non-dominated sorting process in every iteration, making it more expensive compared to one weighted-sum iteration.

5 Results

5.1 Single-snapshot comparison

Results for the one-shot scenario for the fixed seed case are provided in Table 3 below. GWO comes out on top with a weighted score of 13.47, which equates to a data rate of 19.33bps/Hz and Pd equaling 0.760. PSO is not far behind, scoring almost identical in weighted scores but with the best detection probability among all algorithms, 0.777. DE sits in the middle while SA, SFO, and POA sacrifice detection for better rates.

Table 3. Single-snapshot results for the `UMi_NLOS` case at $\alpha = 0.5$.

Solver	Rate (bps/Hz)	P_d	Weighted score
GWO	19.33	0.760	13.47
PSO	18.81	0.777	13.29
DE	17.58	0.697	12.27
SFO	21.34	0.190	11.62
POA	21.33	0.166	11.50
SA	21.00	0.138	11.19

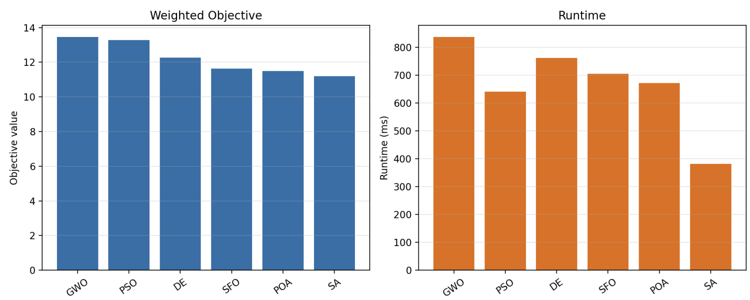

Fig. 2. Single-snapshot comparison in the default detection-probability mode.

The main lesson from the snapshot is that the best throughput methods are not the best detection methods. This is where a detection-probability-centered formulation is useful. Such an approach keeps the optimization from collapsing into a pure communication problem.

Figure 2 shows the same result graphically.

5.2 Dynamic comparison

The dynamic experiment is more informative than the single snapshot because it shows how the methods behave as the channel changes over time. Across 20 CSI updates, GWO again has the highest mean score (9.21) and mean P_d (0.526) under the fixed-seed setup. PSO is a close second at 9.13 and 0.501.

The dynamic study is also useful for interpretation. SA is the fastest method, but it gives the weakest sensing results by a clear margin. GWO and DE take longer, but they preserve much better sensing quality. For offline design, that extra runtime is acceptable. For a stricter real-time setting, PSO may be the more practical compromise because it is close to GWO but cheaper per update.

Figure 3 and Figure 4 show the dynamic summary and the time traces.

5.3 Pareto Analysis and Sensing–Communication Tradeoff

The Pareto study is the clearest example illustrating the main idea in this paper. Using the Grey Wolf Optimizer (GWO), 10 weighted-sum points were generated

Table 4. Dynamic results over 20 CSI updates.

Solver	Mean score	Mean rate	Mean P_d	Mean runtime (ms)
GWO	9.21	13.17	0.526	795.2
PSO	9.13	13.25	0.501	598.7
POA	8.47	13.48	0.346	644.0
DE	8.21	12.58	0.383	755.9
SFO	8.02	13.36	0.268	687.4
SA	6.98	13.17	0.080	395.6

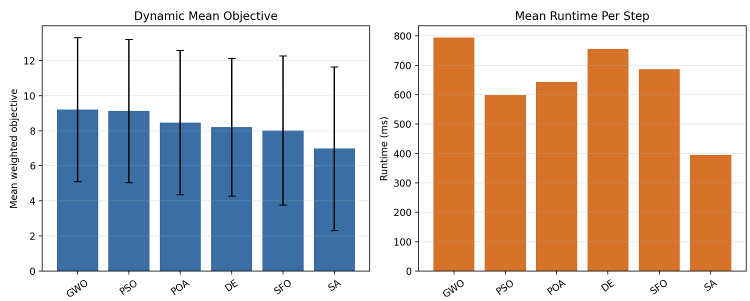

Fig. 3. Mean dynamic performance across the 20 CSI updates.

on $\alpha \in [0, 1]$. As for NSGA-II, 48 non-dominated solutions were obtained, which can be considered a reference for front mapping. However, it should be noted that the two approaches try to solve different problems: GWO tries to find some weighted operating points, whereas NSGA-II tries to map the front. At the sensing-oriented side, GWO reaches $P_d = 0.892$. As for the rate-oriented side, GWO achieves $P_d = 0.212$ and 22.76 bps/Hz. This great range is important since it proves that the extra waveform dimension is meaningful; indeed, the optimizer uses it to cover various parts of the trade-off frontier. Baseline studies also confirm that idea. With equal power, the results are acceptable for the rate but poor for the detection problem. The best results in terms of the rate only are achieved with water filling, which are even worse in terms of the detection problem.

At the sensing-focused end, GWO reaches $P_d = 0.892$. At the rate-focused end, it still keeps $P_d = 0.212$ while reaching 22.76 bps/Hz. That wide span matters because it shows that the added waveform variable is not redundant. The optimizer is using it to move between different parts of the tradeoff curve.

The baselines make the point even clearer. Equal power is decent for rate but weak for detection. Water-filling is the best rate-only baseline, but it is even weaker in P_d. So a classical power-only strategy is not enough if detection probability is the main sensing goal.

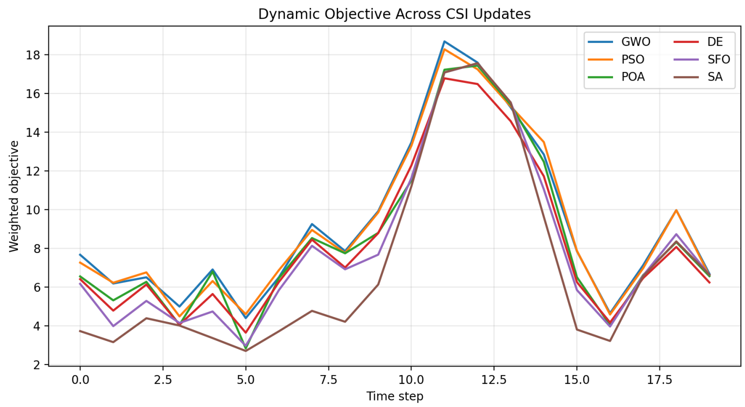

Fig. 4. Objective traces across time for each solver.

Table 5. Key endpoints in the `UMi_NLOS` Pareto study.

Case	Rate (bps/Hz)	P_d	Comment
GWO, $\alpha = 0$	10.70	0.892	sensing-focused end
GWO, $\alpha = 1$	22.76	0.212	rate-focused end
Equal power	20.97	0.139	simple baseline
Water-filling	22.50	0.114	rate-first baseline

5.4 Optimized Waveform Profiles

The optimized waveform profiles are not flat as show in Figure 6. This is important because it depicts the waveform variable is doing real work, rather than acting as a duplicate of the power vector. In practical terms, the solution places more weight on a few subcarriers than others, which helps it balance detection and communication better than a flat profile. The sensing-focused solution ($\alpha = 0$) becomes sharply concentrated, while the rate-focused solution ($\alpha = 1$) is closer to flat but still non-uniform.

The waveform variable should be read as a normalized spectral shaping profile, instead of a full physical beamforming design.

6 Discussion

Taken together, the experiments support three careful conclusions:

1. **The performance obtained by GWO and PSO algorithms is similar while using fixed-seed.** GWO algorithm obtains slightly better performance with regard to the weight function in both dynamic and static experiments; nevertheless, it is not significant enough to state that it has a statistical superiority over PSO algorithm.

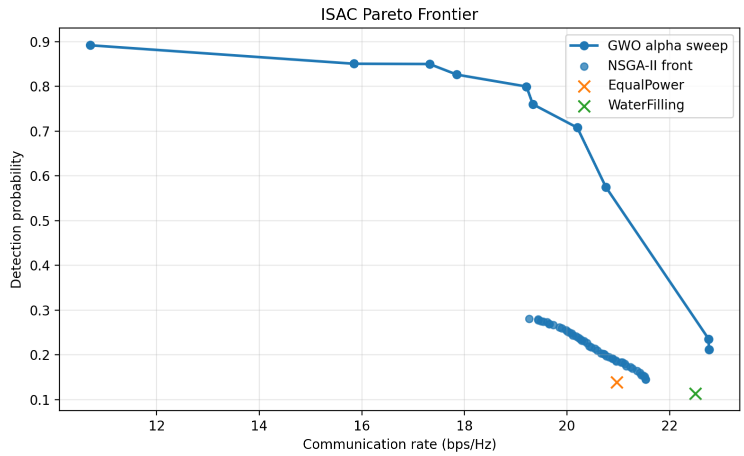

Fig. 5. Pareto-view of the `UMi_NLOS` snapshot.

2. **PSO presents a strong competitor when real-time requirements are essential.** PSO is quite similar to GWO in the total score and slightly better than GWO in snapshot Pd, and there is also a clear indication of higher speed in dynamic P_d.
3. **Waveform co-optimization is of utmost importance.** The non-flat waveform profiles and the wider tradeoff range depict that the added variables change the design space in a meaningful manner.

These findings are specific to the `UMi_NLOS` scenario and the fixed experimental setup and protocol. They have been offered as a reproducible benchmark, rather than general design laws.

7 Limitations and Future Work

This study makes several deliberate simplifications that bound the scope of its conclusions without invalidating the results.

On the experimental side, the static comparison has been based on a single representative channel snapshot rather than a Monte Carlo average over many realizations. Each solver has been further evaluated under one fixed random seed. Consequently, small score differences between the solvers, particularly between GWO and PSO, should be treated as indicative rankings rather than statistically significant results. A multi-seed study with confidence intervals is the natural next step.

On the sensing model side, the system assumes a single point target with a round-trip return and no clutter or multi-target interference. The detection-probability mapping uses an effective integration gain that is calibrated for useful P_d variation across the comparison rather than derived from a physical detector

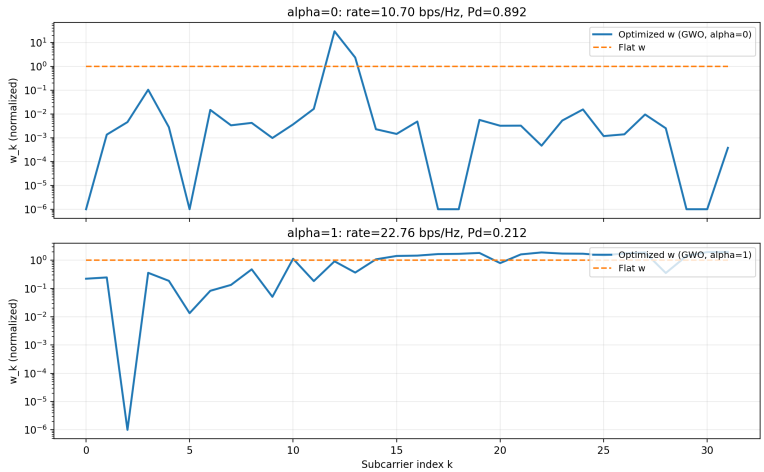

Fig. 6. Example optimized waveform profiles for GWO at the sensing-focused end ($\alpha = 0$) and the rate-focused end ($\alpha = 1$), compared with a flat profile.

design. The paper therefore characterizes detection-aware resource allocation behavior, not a complete radar receiver chain.

Finally, the comparison is limited to metaheuristic search methods operating on the same weighted objective. Detector-side baselines such as matched-filter, Neyman–Pearson, or MMSE beamforming receivers are not included. Extending the benchmark in these directions — through Monte Carlo averaging, richer sensing physics, and detector-level baselines — is left as future work.

8 Conclusion

This paper presents a small but focused ISAC case study for the `UMi_NLOS` scenario. The main message is that detection probability and waveform shaping should be treated as first-class design variables if the sensing task is detection-oriented. Under the fixed-seed protocol used here, GWO has a slight edge on the weighted objective and PSO is a close and often faster alternative; power-only baselines such as water-filling remain strong on rate but weak on detection. The Pareto endpoints and the non-flat optimized waveform profiles support the practical value of detection-aware waveform co-optimization.

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